Impact of Node Density and Propagation Models on Wi-Fi Network Performance Using ns-3

UDP and TCP comparative evaluation of IEEE 802.11g and IEEE 802.11ax WLANs

1,080	2	2	3	360
scheduled ns-3 runs	transport protocols	Wi-Fi standards	propagation models	grouped scenario cells

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Goal: quantify how density, distance, propagation and transport protocol jointly affect reliability, capacity, delay and fairness.

Experimental design and reproducibility

Controlled variables

Transport: UDP and TCP
PHY/MAC: IEEE 802.11g and 802.11ax
Propagation: Friis, LogDistance, Friis–Nakagami
Density: 5, 10, 20, 30 and 50 stations
Distance: 5–50 m; seeds: 1, 2 and 3

Topology

One Access Point (AP)
Multiple wireless stations (STA → AP)
Static nodes; uplink application traffic
FlowMonitor used for packet, byte and delay counters

Run calculation

$$R_{protocol} = 2 \times 3 \times 5 \times 6 \times 3 = 540, \qquad R_{total} = 2 \times 540 = 1080$$

540 runs per protocol

1,080 total scheduled

360 scenario groups

Valid raw outputs

UDP: 536 valid CSV files
TCP: 537 valid CSV files
Failed runs documented as internal ns-3/cppyy segmentation errors and excluded from aggregates

Traceable pipeline

Matrix generation
ns-3 execution per scenario
Raw flow and aggregate CSVs
Seed statistics and IC95
UDP/TCP alignment and deltas
Publication-ready figures and tables

Design | Run | Measure | Analyze | Publish

The simulator links distance and channel conditions to received power, SNR, packet errors and observed delivery.

Wavelength

$$\lambda = \frac{c}{f_c}$$

Friis received power

$$P_r(d) = P_t G_t G_r \left(\frac{\lambda}{4\pi d}\right)^2$$

Log-Distance path loss

$$PL(d) = PL(d_0) + 10n \log_{10}\left(\frac{d}{d_0}\right)$$

Nakagami fading envelope

$$f_R(r) = \frac{2m^m}{\Gamma(m)\Omega^m} r^{2m-1} e^{-\frac{mr^2}{\Omega}}, \quad r \geq 0$$

SNR

$$\gamma = \frac{P_r}{N_0 B}$$

Packet error link

$$PER = 1 - (1 - BER)^L$$

Physical interpretation

Propagation loss → ↓ received power → ↓ SNR → ↑ frame errors → ↓ PDR → ↓ useful throughput

Why three propagation models?

Friis: optimistic free-space reference
LogDistance: stronger deterministic attenuation
Friis–Nakagami: path loss plus multipath fading

Why compare 802.11g and 802.11ax?

802.11g is a legacy OFDM baseline
802.11ax offers a higher-capacity PHY/MAC generation
Both are stressed by the same density and distance grid

Key caution

High TCP PDR reflects transport recovery
It does not imply that the wireless channel is loss-free

Metrics used to quantify performance

The analysis combines reliability, capacity, latency, load, fairness and statistical uncertainty.

Throughput

$$T = \frac{8B_{rx}}{t_{last\ rx} - t_{first\ tx}} \times 10^{-6}\ \mathrm{Mbps}$$

PDR and packet loss

$$PDR = \frac{N_{rx}}{N_{tx}}, \quad PLR = 1 - PDR$$

Mean delay

$$\bar{D} = \frac{\sum_{k=1}^{N_{rx}} D_k}{N_{rx}}$$

Load, per-node capacity and efficiency

$$L_{off} = N_{STA} R_{app}, \quad T_{node} = \frac{T_{agg}}{N_{STA}}, \quad \eta = \frac{T_{agg}}{L_{off}}$$

Jain fairness

$$J = \frac{\left(\sum_{i=1}^{N} x_i\right)^2}{N \sum_{i=1}^{N} x_i^2}, \quad 0 < J \leq 1$$

95% confidence interval

$$CI_{95\%} = \bar{x} \pm t_{0.975,n-1} \frac{s}{\sqrt{n}}$$

Reliability

PDR measures delivered packets over transmitted packets.
UDP exposes raw loss because it does not retransmit.

Capacity and fairness

Throughput measures useful delivered rate.
Fairness evaluates how equally flows share capacity.

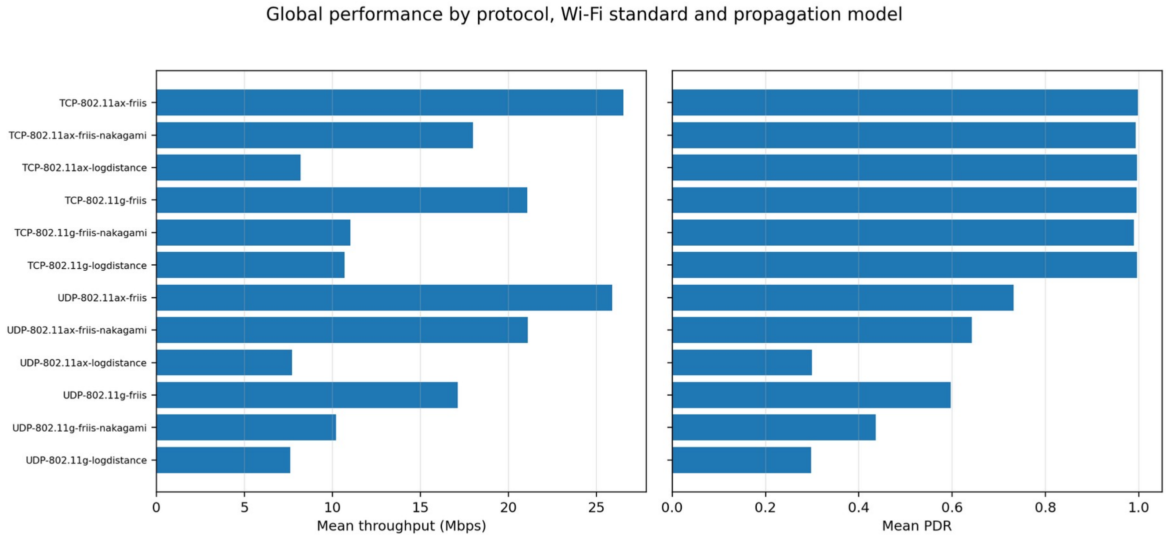
Uncertainty

Seeds give independent observations per scenario.
IC95 quantifies uncertainty around seed-based means.

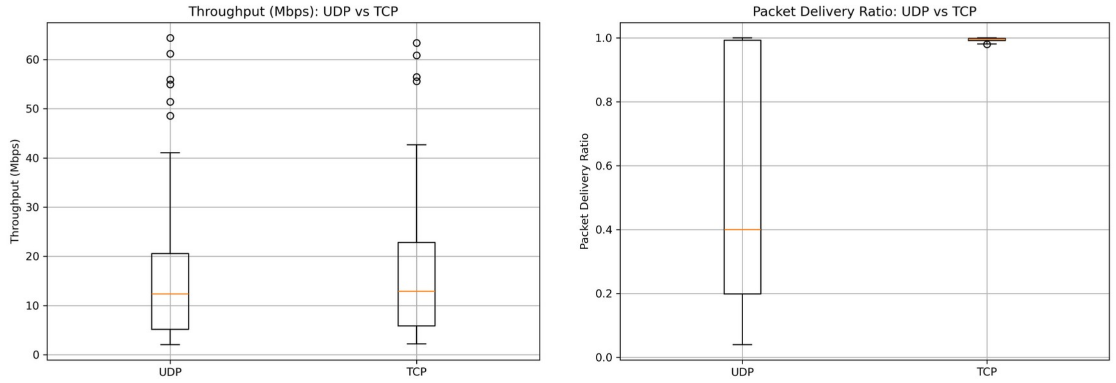
Final interpretation always combines PDR + throughput + delay + fairness.

Main comparative results: UDP vs TCP

Global results show that transport recovery changes delivery reliability, while capacity remains constrained by PHY, density and propagation.



Mean throughput and PDR by protocol, standard and propagation model.



Throughput and PDR distributions across all scenario groups.

Observed patterns

TCP PDR is close to unity due to acknowledgements and retransmissions.
UDP PDR varies widely because losses are not recovered.

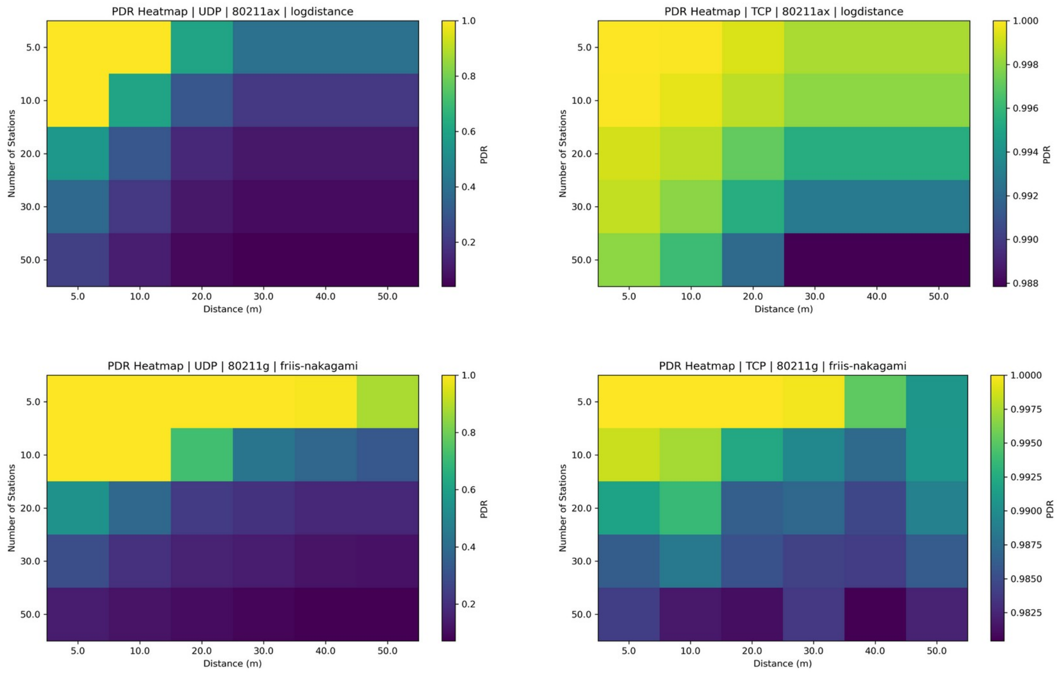
Highest mean throughput

802.11ax + Friis gives the highest capacity region.
TCP ≈ 26.5 Mbps; UDP ≈ 25.9 Mbps in the global summary.

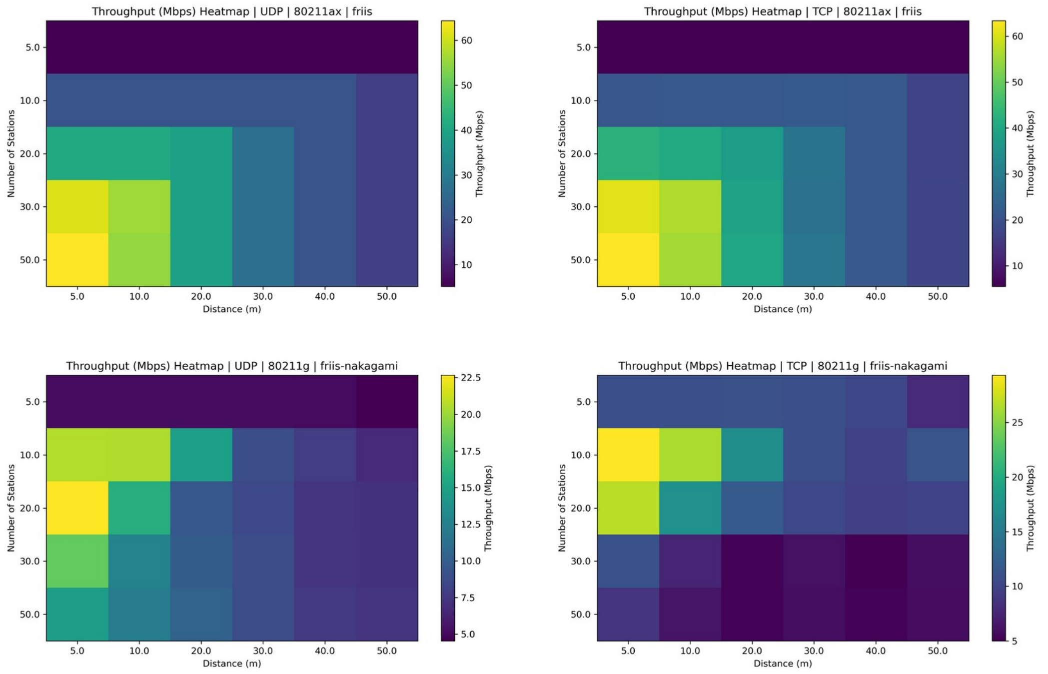
Key interpretation

High TCP PDR ≠ better radio channel.
It means transport-layer recovery masked part of wireless loss.

Density, distance and final conclusions



PDR heatmaps: reliability across distance and station density.



Throughput heatmaps: capacity under favourable and fading conditions.

Reliability

UDP falls sharply with LogDistance, density and distance.
TCP preserves effective delivery but may reduce sending pressure.

Scalability

More stations increase contention for the same medium.
Throughput per station and Jain fairness decline with density.

Conclusion

802.11ax improves the capacity ceiling, but propagation and MAC contention still dominate.
Design decisions must jointly consider throughput, PDR, delay and fairness.

Thanks for your attention

Questions?